

# Report Title: Replace later

Computational Cognitive Science – Final Project, Summer Term 2026

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*Requested assessment: Full grading*

**Abstract** Here comes the abstract.

## 1 INTRODUCTION

The present study examines probability discounting by giving the participants choices between a certain and an uncertain outcome that vary in probability and magnitude. Such choice tasks provide a way to study how objective probabilities are transformed into subjective values that guide decisions. Useful in this context are computational models because they give a distinction of the valuation process from the subsequent choice process. Instead of assuming a direct dependence of the choices from probabilities, they estimate the subjective value of each option and then link them to the observed behavior. This approach is here extended by also modelling the participants' reaction times (RTs), assuming that both RT and choice arise from the same subjective values. Modelling choice and RT jointly provides an opportunity to investigate whether RT contains information about the valuation process that is not captured by choices alone.

RT is rarely incorporated into models of discounting because RT is influenced by more than just valuation, for example processing speed. That means linking RT to a valuation model requires additional assumptions about which aspects are reflected in RT. Nevertheless, RT may provide useful information about the decision difficulty: When the subjective value difference is high, the decision may be easier and require less time than with a low value difference.

The valuation model used throughout the study is a hyperbolic probability discounting function, choice is linked to the value difference using a logistic choice rule. The valuation and choice components of the model are fixed across the analyses. The primary modelling question is how the same value differences are translated into RT. Therefore three competing hypotheses are compared: The first is an absolute conflict model which assumes that RT reflects the absolute degree of conflict between the choices, the second one, a squared conflict model, allows the effect of the value difference on RT to be nonlinear, the third one is an exponential mapping of the value difference to RT. Due to identical RT distribution and the difference in their value-to-RT mapping these hypotheses can be compared directly.

This report addresses three main questions: First, which value-to-RT mapping provides the best account of RT behavior when tested on independent data? Second, are the parameters of the different models identifiable and reliable? And lastly, does modelling RT improve the prediction of choices relative to a choice-only model? Together these analyses investigate whether RT provides additional information about the valuation process beyond the information in choices.

## 2 METHODS

### 2.1 Data

The dataset consists of  $N = 49$  participants who completed two 200-trial runs of a probability discounting task in which each trial presented a choice between a certain ( $p = 1$ ), and an uncertain outcome with probability  $p$ , varying in outcome magnitudes and probability. The task included reward and loss conditions. The models used the certain outcome magnitude  $r_{\text{cert}}$ , the uncertain outcome one  $r_{\text{uncert}}$  and the probability  $p$  to determine subjective value. Choice was coded as a binary variable as 1 for the certain and 0 for the uncertain choice, and RT was measured in milliseconds.

Before modelling, choices were recoded accordingly and probabilities expressed as percentages were converted to  $[0,1]$ . Missing choice and RT observations occurred on the same trials. There were 51 incomplete trials in

Run A and 17 in Run B. Trials with missing or invalid choices or invalid RT were excluded from the likelihood. No non-positive RTs were observed and no RT outlier cutoff was applied. Behavioral descriptives are summarized in table 1, separately for run and condition. RT distributions are shown in figure 1. Mean RTs and mean certain choice proportions were higher for losses than rewards in both runs.

## 2.2 Models

All models share the same hyperbolic probability-discounting valuation model. For trial  $i$ , the subjective values  $v_{\text{uncert}}$  and  $v_{\text{cert}}$  of the uncertain and the certain option, respectively, were defined as:

$$v_{\text{uncert},i} = \frac{r_{\text{uncert},i}}{1 + h \frac{1-p_i}{p_i}}, \quad v_{\text{cert},i} = r_{\text{cert},i} \quad (1)$$

with  $h \geq 0$  being a discounting parameter. The subjective value difference  $\Delta v$  was defined as  $\Delta v_i = v_{\text{cert},i} - v_{\text{uncert},i}$  and entered a logistic choice rule:

$$\pi_i = \frac{1}{1 + \exp(-\beta \Delta v_i)} \quad (2)$$

where  $\beta \geq 0$  controls choice sensitivity. For an observed choice  $y_i \in \{0, 1\}$ , the choice likelihood was  $p(y_i | \Delta v_i, \beta) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$ .

**Psychological hypotheses.** The three competing RT models differed in the function that maps  $\Delta v$  onto the location parameter  $\mu_i$  of the RT distribution:

$$\text{Absolute: } \mu_i = a - b|\Delta v_i|, \text{ squared: } \mu_i = a - b(\Delta v_i)^2, \text{ exponential: } \mu_i = a + b \exp[-(\Delta v_i)^2]. \quad (3)$$

All three mappings assume that smaller  $\Delta v$  produce higher decision difficulty and therefore longer RTs (Berlyne, 1957), but differ in the functional form of this relationship. The absolute and squared mappings represent linear and nonlinear measures of value conflict, respectively, whereas the exponential mapping assumes a bounded effect that decreases as  $\Delta v$  become larger (Koppe, 2026). Here,  $a$  is the baseline log-RT and  $b \geq 0$  controls the strength of the value difference effect.

RTs followed the same shifted log-normal distribution in all three models. The trial-specific RT likelihood was

$$p(\text{RT}_i | \mu_i, \sigma, \tau) = \frac{1}{(\text{RT}_i - \tau)\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log(\text{RT}_i - \tau) - \mu_i)^2}{2\sigma^2}\right], \quad (4)$$

for  $\text{RT}_i > \tau$ , where  $\sigma > 0$  controls RT variability and  $\tau \geq 0$  is a lower-bound shift. Assuming conditional independence of choice and RT given the model parameters and subjective values, the joint trial likelihood was

$$p(y_i, \text{RT}_i | \theta) = p(y_i | \Delta v_i, \beta) p(\text{RT}_i | \mu_i, \sigma, \tau), \quad \theta = (\beta, h, a, b, \sigma, \tau). \quad (5)$$

**Choice-only baseline.** A choice-only baseline used the same valuation and choice models, but left out the RT likelihood. Its parameter vector was  $\theta_{\text{choice}} = (\beta, h)$ , with the choice likelihood as its objective. This provided a baseline for testing whether incorporating RT improves the choice modelling.

## 2.3 Inference

Parameters were estimated independently for each participant using maximum likelihood estimation (MLE). For the choice-only model, the choice negative log-likelihood was minimized with respect to  $\beta$  and  $h$ , whereas for the three full choice-RT models, the joint negative log-likelihood of choice and reaction time was minimized with respect to

$$\theta = (\beta, h, a, b, \sigma, \tau).$$

No priors, regularization, or hierarchical pooling were used. Each model specifies a complete likelihood for the observed data. Estimation was performed separately at the participant level to obtain participant-specific parameter estimates for the subsequent test-retest reliability analysis.

All models were optimized using the L-BFGS-B algorithm, which allows parameter bounds to be imposed directly. The constraints were

$$\beta \geq 0, \quad 0 \leq h \leq 100, \quad a \in \mathbb{R}, \quad b \geq 0, \quad \sigma \geq 10^{-6}, \quad 0 \leq \tau \leq \min(RT) - 10^{-6}.$$

The finite upper bound on  $h$  was used to restrict optimization in extreme regions with poor practical identifiability and to improve numerical stability. The participant-specific upper bound on  $\tau$  ensured that  $RT - \tau > 0$  for all trials. Multi-start optimization was used to reduce sensitivity to starting values and local optima. The choice-only model was initialized from a broad grid of  $\beta$  and  $h$  values spanning several orders of magnitude. For the full RT models, the choice-only estimates were used as the center of local starting points, while initial values for  $a$ ,  $b$ , and  $\sigma$  were informed by the observed relationship between shifted log-RT and the model-specific value-difference predictor. These data-informed values were used only for initialization; all parameters were subsequently estimated jointly by MLE. Additional global anchors were included to explore regions farther from the choice-only solution.

To improve numerical stability,  $\tau$  was internally rescaled relative to the participant-specific bound  $\min(RT) - 10^{-6}$  and optimized on the interval  $[0, 1]$ , before being transformed back to milliseconds after fitting. This was a numerical reparameterization only and did not alter the likelihood model. For the full RT models, stricter convergence tolerances and a maximum of 2,000 iterations were used. An optimization run was considered valid only if the optimizer reported success and both the objective value and all parameter estimates were finite. Among the valid solutions obtained from multiple starting points, the solution with the lowest negative log-likelihood was retained as the final estimate. All primary parameter estimates were obtained independently from each participant’s Run A data. The same fitting procedure was applied independently to Run B only for the test-retest reliability analysis.

## 2.4 Evaluation Plan

Parameter recovery was assessed by generating synthetic data from known true parameter values and re-estimating the corresponding full model using the same fitting procedure applied to the empirical data. For each RT mapping, 40 generating parameter sets were sampled from model-specific, empirically informed ranges, and three simulations were performed per set while holding the empirical 200-trial Run A task design fixed. Because plausible values of  $\beta$  and  $h$  span several orders of magnitude, they were sampled on a log scale, whereas  $a$ ,  $b$ ,  $\sigma$ , and  $\tau$  were sampled uniformly from realistic ranges. Recovery of the six full-model parameters was evaluated using bias, RMSE, and the correlation between true and recovered values. To assess whether including RT improved identifiability of the shared choice parameters, recovery of  $\beta$  and  $h$  was additionally compared between the full and choice-only models using RMSE and MAE. Because the three repetitions generated from the same parameter set were not independent, absolute recovery errors were first averaged at the parameter-set level. Full- and choice-only absolute recovery errors were then compared across the 40 independent parameter sets using paired Wilcoxon signed-rank tests with Holm correction.

Test-retest reliability was assessed by fitting Run A and Run B independently for each participant and correlating the resulting parameter estimates across participants. Pearson correlation was used as the primary measure of cross-run stability, with Spearman rank correlation additionally calculated to assess robustness to extreme or boundary estimates. For the shared parameters  $\beta$  and  $h$ , changes in reliability between the full and choice-only models were compared descriptively using differences in cross-run correlations.

For out-of-sample prediction, only parameter estimates obtained from Run A were used and held fixed when predicting Run B; no refitting was performed on Run B. Reward and loss trials were evaluated separately. Choice prediction was assessed using participant- and condition-level mean predictive choice log-likelihood per trial as the primary metric, with accuracy as a supplementary metric. RT prediction was evaluated using participant- and condition-level mean predictive RT log-likelihood per trial as the primary metric and predictive-median MAE as a supplementary point-prediction measure. Because the shifted log-normal density is defined only for  $RT > \tau$ , RT support coverage was reported separately. Support-violating trials were retained when reporting coverage but excluded only from the relative RT log-likelihood comparison, which was restricted to common-support trials for which all three models had valid support. Predictive-median MAE was calculated across all held-out Run B trials regardless of support.

The three RT mappings were compared based on participant-level mean Run B RT predictive log-likelihood per trial. Friedman omnibus tests were conducted separately for the reward and loss conditions, followed by pairwise

Wilcoxon signed-rank tests with Holm correction within each condition. The contribution of RT to choice modeling was assessed separately by comparing each full model with the choice-only model on the same held-out Run B choice predictive log-likelihood, with accuracy differences examined as a supplementary measure. Participant-level log-likelihood differences were tested within each condition using paired Wilcoxon signed-rank tests with Holm correction across the six model-by-condition comparisons. Joint choice-RT likelihoods were not directly compared with choice-only likelihoods because they are based on different observed data modalities.

### 3 RESULTS

#### 3.1 Parameter Recovery and Identifiability

Recovery plots showing true versus recovered parameter values are presented in Figures 2-4. The identity lines in the plots represent perfect recovery. The parameter recovery results indicate that the RT models differed in how well their parameters could be identified (Table 2).  $\beta$  was generally well recovered for all RT models, with high true-recovered correlations ( $r = 0.94$ - $0.95$ ).  $h$  had moderate to high correlations for the absolute and squared models ( $r = 0.84$  and  $r = 0.81$ ), while recovery was noticeably weaker for the exponential model ( $r = 0.52$ ). The RT parameters also varied in recoverability:  $a$  was consistently well recovered across all models ( $r > 0.95$ ), while  $b$  showed weak recovery ( $r = 0.43$ - $0.49$ ).  $\sigma$  recovery was good for the absolute and exponential models and weaker for the squared model. The parameter  $\tau$  showed moderate recovery in all models.

The inclusion of RT reduced aggregate recovery error for  $h$ , particularly for the absolute and squared conflict models (Tables 3-4). The largest improvements were obtained for the absolute and squared conflict models, where RMSE decreased from 8.68 to 1.88 and from 8.62 to 2.57, respectively. For the exponential model the improvement was only 0.04. However, these improvements were less consistent across the 40 independent generating parameter sets, with relatively small median improvements. In contrast, recovery of  $\beta$  changed little with the inclusion of RT.

#### 3.2 Reliability

Table 5 summarizes the Pearson test-retest reliability results. For the choice-only model,  $\beta$  showed high Pearson test-retest reliability ( $r = 0.825$ ), whereas  $h$  showed almost no linear cross-run association ( $r = -0.001$ ). Across the three RT models,  $\beta$  showed consistently high Pearson reliability ( $r = 0.823$ - $0.826$ ). Among the remaining full-model parameters,  $a$  showed relatively high cross-run correlations and  $\sigma$  showed moderate correlations, whereas the reliability of  $b$ ,  $h$ , and  $\tau$  varied more strongly across the RT mappings (Table 5).

Participant-level Run A and Run B parameter estimates are shown in Figures 5-8 with one point per participant. For the shared parameters, adding RT had little effect on the reliability of  $\beta$ . Pearson reliability of  $h$  was numerically higher in all three full RT models than in the choice-only baseline. In the Spearman robustness analysis, however,  $h$  reliability was lower for the absolute model but higher for the squared and exponential models than for the choice-only baseline. Thus, the squared and exponential mappings showed the most consistent descriptive evidence of increased cross-run stability for  $h$  across both correlation measures. Because differences between reliability coefficients were not formally tested, these comparisons are descriptive rather than evidence of statistically significant improvements (Table 6).

#### 3.3 Predictive Performance (Choice and RT, by Condition)

Held-out choice performance was similar across models. Mean choice accuracy ranged from 0.631 to 0.650 in the loss condition and from 0.649 to 0.663 in the reward condition. Mean predictive choice log-likelihood also showed no consistent advantage of the full RT models over the choice-only baseline (Table 7). Paired Wilcoxon signed-rank tests comparing each full model with the choice-only baseline showed no significant improvement in predictive choice log-likelihood after Holm correction across the six model-condition comparisons (Table 8). Thus, including RT did not reliably improve held-out choice prediction relative to the choice-only baseline (Table 8; Figure 9, left).

RT support coverage was high and identical across the three mappings: 99.73% of reward trials and 99.84% of loss trials satisfied the shifted log-normal support condition. Median RT MAE differences were comparatively

small. For reward trials, median MAEs were 806, 800, and 805 ms for the absolute, squared, and exponential models, respectively; for loss trials, they were 816, 816, and 797 ms. On common-support trials, median participant-level RT predictive log-likelihood was highest for the exponential model in both conditions. For reward trials, median log-likelihoods were -8.294, -8.316, and -8.252 for the absolute, squared, and exponential models, respectively. For loss trials, the corresponding values were -8.371, -8.379, and -8.296.

Friedman tests indicated differences among the RT mappings for both reward ( $\chi^2_F(2) = 12.29, p = 0.002$ ) and loss ( $\chi^2_F(2) = 7.88, p = 0.019$ ). Holm-corrected pairwise Wilcoxon tests showed that the exponential model had higher predictive log-likelihood than the absolute model in reward ( $p_{\text{Holm}} < 0.001$ ) and loss ( $p_{\text{Holm}} = 0.024$ ), and than the squared model in reward ( $p_{\text{Holm}} = 0.001$ ) and loss ( $p_{\text{Holm}} = 0.015$ ). The absolute and squared models did not differ reliably.

All three models also tended to overpredict mean RT. For example, for the absolute model, the observed RT mean was approximately 2060 ms in the loss condition and 1854 ms in the reward condition, whereas the predicted mean RTs were around 2600 ms. This indicates a shared upward prediction bias rather than a clearly model-specific effect (Tables 9-11; Figure 9, right).

Overall, the full RT models did not reliably improve held-out choice prediction relative to the choice-only baseline, whereas the RT mappings differed in predictive log-likelihood, with the exponential mapping showing the strongest held-out RT prediction.

### 3.4 Value of the RT Modality

RT information did not lead to a consistent improvement in held-out choice prediction. Participant-level differences in Run B choice log-likelihood between the full RT models and the choice-only baseline were typically close to zero in both reward and loss conditions, and none of the comparisons remained significant after Holm correction. Accuracy differences were also small, with a median difference of zero for every model-condition comparison. Thus, including RT information during fitting did not provide a clear benefit for subsequent choice prediction (Table 12, Panel A).

The effect of RT information on parameter recovery was parameter-specific. Recovery of  $\beta$  changed little relative to the choice-only baseline. In contrast, aggregate recovery error for  $h$  decreased substantially for the Absolute and Squared models, while remaining nearly unchanged for the Exponential model. However, improvements at the level of independent generating parameter sets were small and not consistent across all sets. The reduction in aggregate RMSE therefore does not necessarily indicate a general improvement in  $h$  identifiability, but may instead reflect a reduction in some large recovery errors (Table 12, Panel B).

A similarly selective pattern emerged for test-retest reliability. Cross-run reliability of  $\beta$  remained essentially unchanged after including RT information. For  $h$ , Pearson reliability was higher for all full RT models than for the choice-only baseline. However, when the Spearman robustness analysis was also considered, both correlation measures were higher than the choice-only baseline only for the Squared and Exponential models. Because no formal statistical test was conducted on differences between reliability coefficients, these results should be interpreted as descriptive evidence of stronger cross-run stability rather than as a statistically significant improvement (Table 12, Panel C).

Overall, the added value of RT was selective rather than reflecting a general improvement of the shared choice model. RT information provided little additional benefit for estimating  $\beta$  or for predicting held-out choices, but it may have helped reduce large recovery errors and improve the cross-run stability of  $h$  for some RT mappings. These limited benefits must be weighed against the greater model complexity and the additional assumptions required to model the RT-generating process.

## 4 DISCUSSION

## 5 REFERENCES

- [1] Berlyne, D. E. (1957). Conflict and choice time. *British Journal of Psychology*, 48(2), 106-118.
- [2] Prof. Dr. Georgia Koppe (2026). Computational Cognitive Science. Interdisziplinäres Zentrum für Wissenschaftliches Rechnen, Universität Heidelberg.

## 6 APPENDIX

Table 1: Behavioral descriptive statistics by run and condition.

| Run   | Condition | Mean Valid Trials | Mean Certain Choice Proportion | Mean RT     | Median RT |
|-------|-----------|-------------------|--------------------------------|-------------|-----------|
| Run A | Loss      | 99.183673         | 0.614329                       | 2737.264323 | 2450.5    |
| Run A | Reward    | 99.775510         | 0.597717                       | 2299.871598 | 2038.9    |
| Run B | Loss      | 99.775510         | 0.519980                       | 2060.173486 | 1833.9    |
| Run B | Reward    | 99.877551         | 0.474191                       | 1854.191345 | 1708.5    |

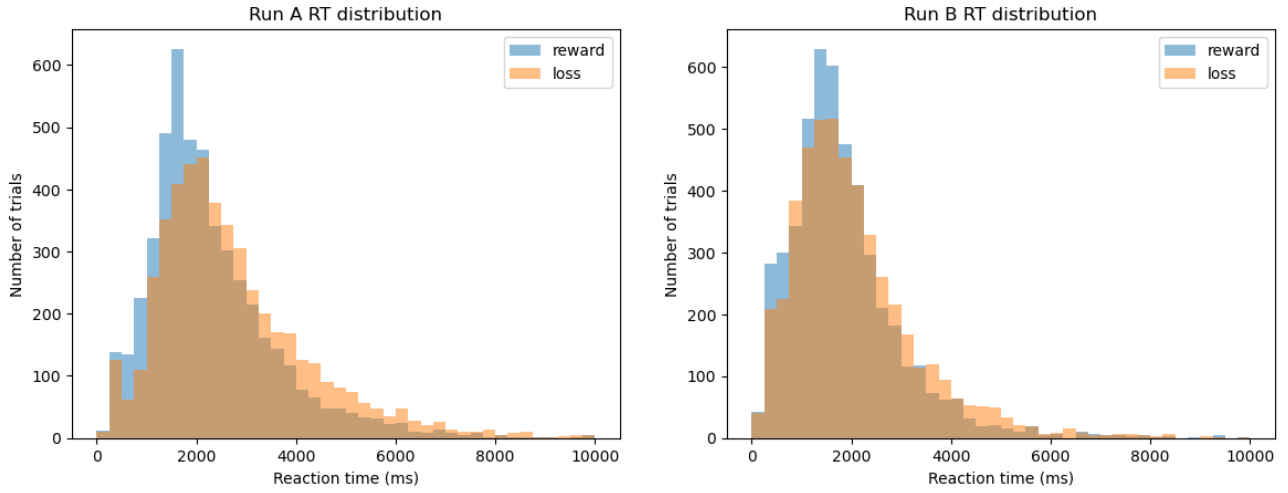


Figure 1: RT distribution for Run A and Run B separately for reward and loss trials.

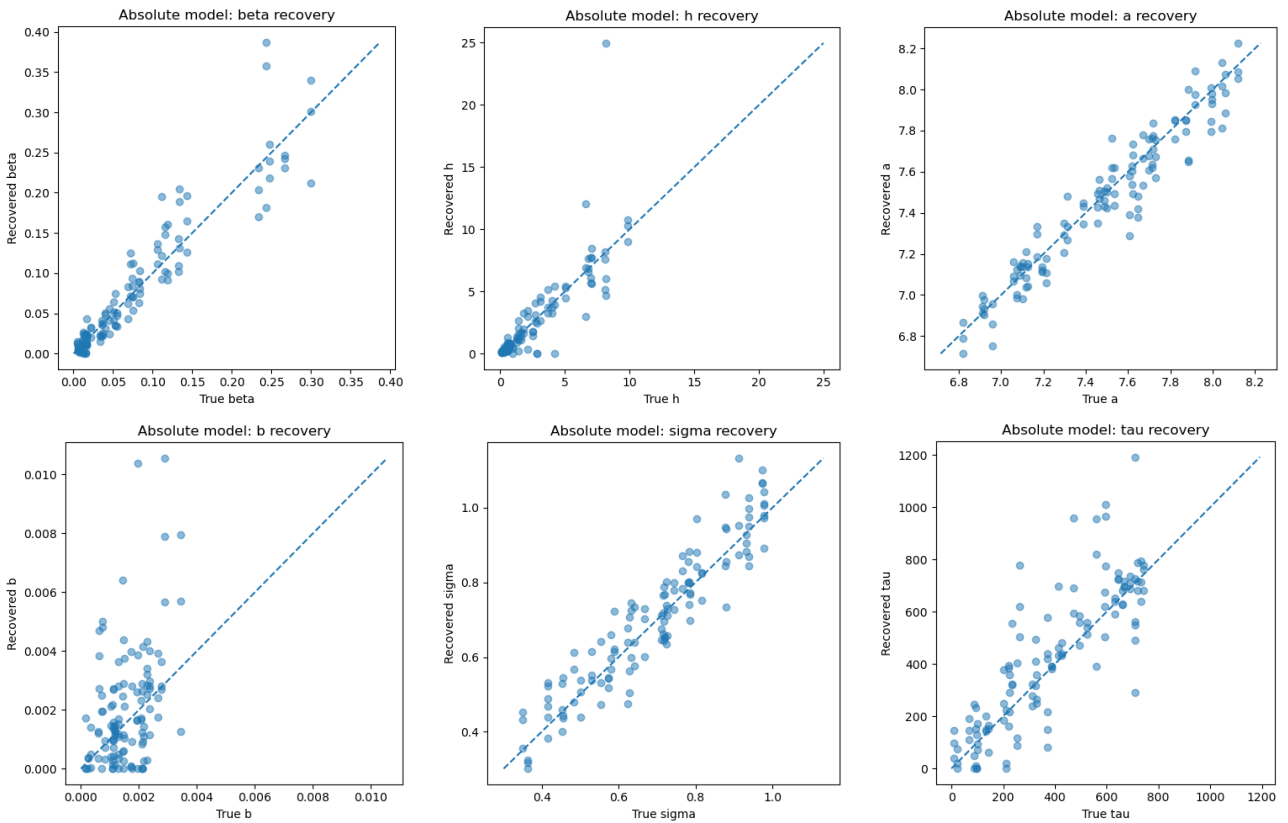


Figure 2: Absolute conflict model parameter recovery plots.

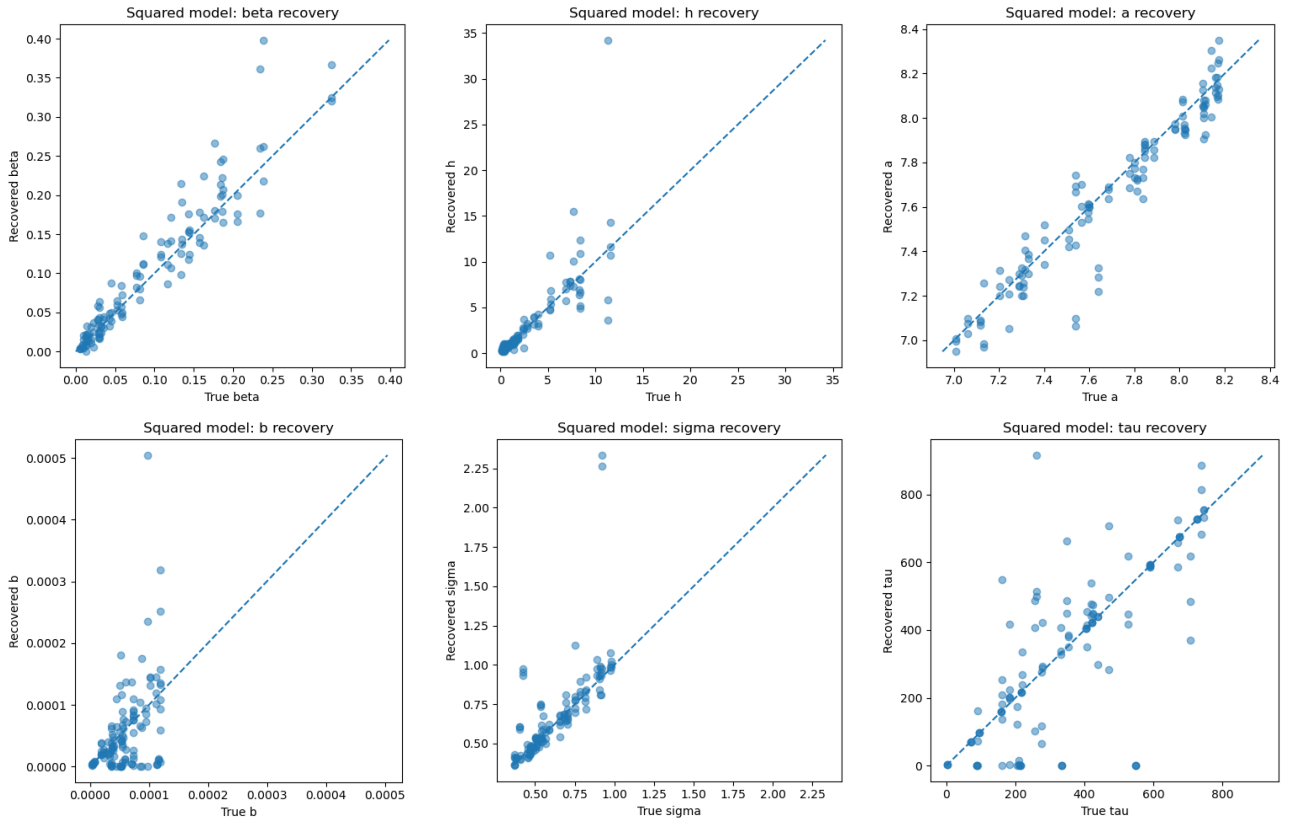


Figure 3: Squared conflict model parameter recovery plots.

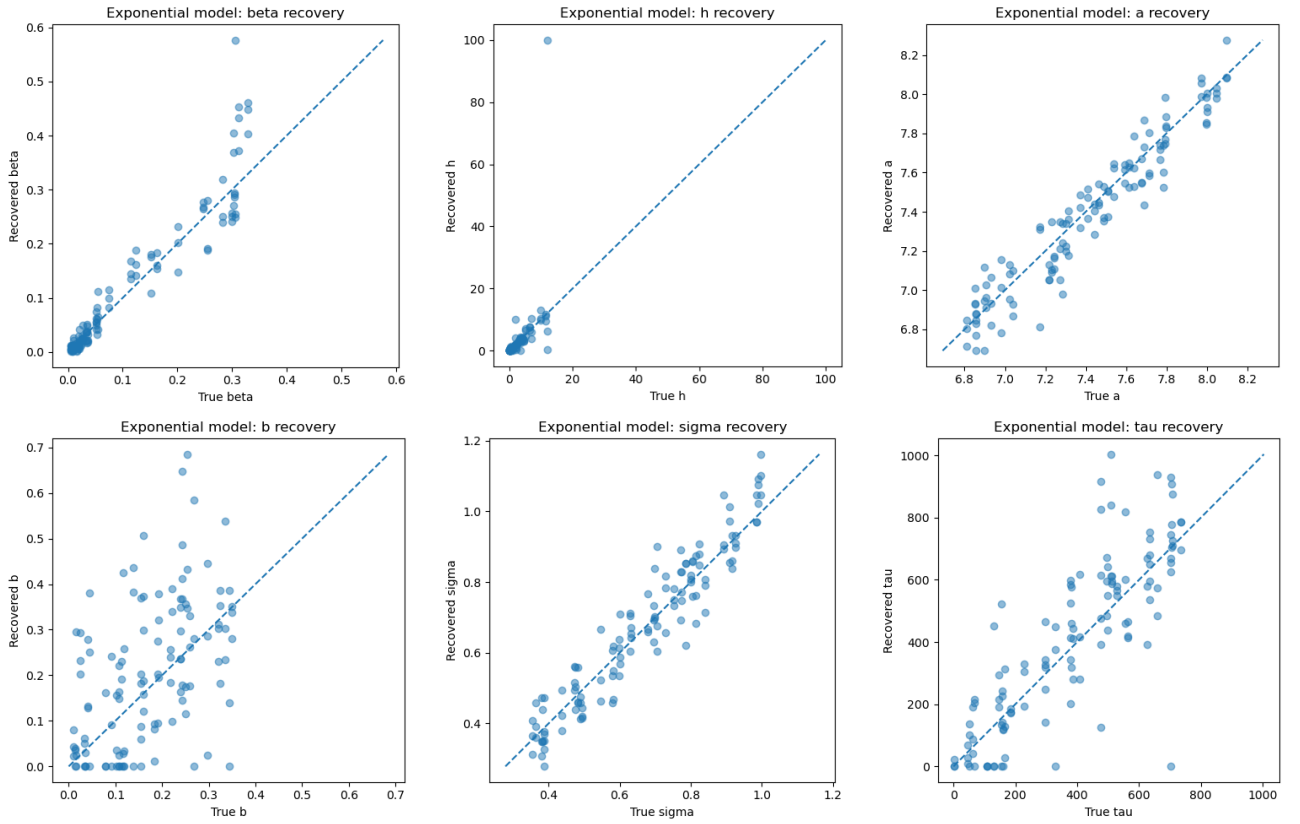


Figure 4: Exponential model parameter recovery plots.

Table 2: Parameter recovery summary.

| Model             | Parameter | Bias       | RMSE       | Correlation |
|-------------------|-----------|------------|------------|-------------|
| Absolute conflict | $\beta$   | 0.002693   | 0.028661   | 0.941256    |
|                   | $h$       | 0.014277   | 1.875806   | 0.838124    |
|                   | $a$       | -0.021145  | 0.100893   | 0.961849    |
|                   | $b$       | 0.000453   | 0.001857   | 0.425069    |
|                   | $\sigma$  | 0.015425   | 0.069458   | 0.934791    |
|                   | $\tau$    | 40.194861  | 153.165480 | 0.838171    |
| Squared conflict  | $\beta$   | 0.009328   | 0.030307   | 0.949655    |
|                   | $h$       | 0.217057   | 2.566000   | 0.814112    |
|                   | $a$       | -0.032828  | 0.114343   | 0.956362    |
|                   | $b$       | -0.000004  | 0.000063   | 0.459258    |
|                   | $\sigma$  | 0.053629   | 0.209938   | 0.714162    |
|                   | $\tau$    | -19.684938 | 161.725961 | 0.764588    |
| Exponential       | $\beta$   | 0.008284   | 0.043213   | 0.950002    |
|                   | $h$       | 0.637440   | 8.197393   | 0.524626    |
|                   | $a$       | -0.021936  | 0.112668   | 0.957746    |
|                   | $b$       | 0.032344   | 0.144445   | 0.494439    |
|                   | $\sigma$  | 0.007824   | 0.067391   | 0.950970    |
|                   | $\tau$    | 25.321610  | 155.602488 | 0.826868    |

Table 3: Choice-only versus RT model recovery data for identifiability improvement analysis.

| Model             | Parameter | Full RMSE | Choice RMSE | RMSE Improvement | Full MAE | Choice MAE | MAE Improvement |
|-------------------|-----------|-----------|-------------|------------------|----------|------------|-----------------|
| Absolute conflict | $\beta$   | 0.028661  | 0.028486    | -0.000175        | 0.018021 | 0.018316   | 0.000294        |
|                   | $h$       | 1.875806  | 8.675386    | 6.799580         | 0.726613 | 1.447669   | 0.721056        |
| Squared conflict  | $\beta$   | 0.030307  | 0.027261    | -0.003046        | 0.019006 | 0.017489   | -0.001517       |
|                   | $h$       | 2.566000  | 8.620525    | 6.054525         | 0.854353 | 1.621860   | 0.767506        |
| Exponential       | $\beta$   | 0.043213  | 0.043516    | 0.000303         | 0.023991 | 0.024336   | 0.000345        |
|                   | $h$       | 8.197393  | 8.239674    | 0.042281         | 1.376429 | 1.530502   | 0.154074        |

Table 4: Recovery improvement comparison across the 40 true parameter sets.

| Model             | Parameter | $N$ | Full better | Choice better | Ties | Median improvement | Mean improvement |
|-------------------|-----------|-----|-------------|---------------|------|--------------------|------------------|
| Absolute conflict | $\beta$   | 40  | 23          | 17            | 0    | 0.000110           | 0.000294         |
|                   | $h$       | 40  | 26          | 14            | 0    | 0.013071           | 0.721056         |
| Squared conflict  | $\beta$   | 40  | 22          | 18            | 0    | 0.000046           | -0.001517        |
|                   | $h$       | 40  | 22          | 18            | 0    | 0.016477           | 0.767506         |
| Exponential       | $\beta$   | 40  | 24          | 16            | 0    | 0.000398           | 0.000345         |
|                   | $h$       | 40  | 25          | 15            | 0    | 0.005523           | 0.154074         |



Table 5: Pearson test-retest reliability summary.

| Model             | Parameter | $N$ | Pearson $r$ | $p$ -value                 |
|-------------------|-----------|-----|-------------|----------------------------|
| Choice-only       | $\beta$   | 49  | 0.824722    | $3.227708 \times 10^{-13}$ |
|                   | $h$       | 49  | -0.001409   | $9.923337 \times 10^{-1}$  |
| Absolute conflict | $\beta$   | 49  | 0.825995    | $2.760734 \times 10^{-13}$ |
|                   | $h$       | 49  | 0.288386    | $4.448321 \times 10^{-2}$  |
|                   | $a$       | 49  | 0.747432    | $6.847759 \times 10^{-10}$ |
|                   | $b$       | 49  | 0.032928    | $8.222833 \times 10^{-1}$  |
|                   | $\sigma$  | 49  | 0.658965    | $2.635005 \times 10^{-7}$  |
|                   | $\tau$    | 49  | 0.394360    | $5.049201 \times 10^{-3}$  |
| Squared conflict  | $\beta$   | 49  | 0.826455    | $2.608256 \times 10^{-13}$ |
|                   | $h$       | 49  | 0.464414    | $7.751620 \times 10^{-4}$  |
|                   | $a$       | 49  | 0.745760    | $7.834996 \times 10^{-10}$ |
|                   | $b$       | 49  | -0.080150   | $5.840716 \times 10^{-1}$  |
|                   | $\sigma$  | 49  | 0.663860    | $1.996832 \times 10^{-7}$  |
|                   | $\tau$    | 49  | 0.390692    | $5.511802 \times 10^{-3}$  |
| Exponential       | $\beta$   | 49  | 0.823026    | $3.967691 \times 10^{-13}$ |
|                   | $h$       | 49  | 0.457842    | $9.402719 \times 10^{-4}$  |
|                   | $a$       | 49  | 0.755430    | $3.543417 \times 10^{-10}$ |
|                   | $b$       | 49  | 0.357331    | $1.171651 \times 10^{-2}$  |
|                   | $\sigma$  | 49  | 0.660066    | $2.476772 \times 10^{-7}$  |
|                   | $\tau$    | 49  | 0.394154    | $5.074247 \times 10^{-3}$  |

Table 6: Test-retest reliability comparison for the shared parameters. Pearson correlation was the primary reliability measure, with Spearman correlation reported as a robustness analysis.

| Parameter | Measure         | Choice-only | Absolute | Squared  | Exponential |
|-----------|-----------------|-------------|----------|----------|-------------|
| $\beta$   | Pearson $r$     | 0.824722    | 0.825995 | 0.826455 | 0.823026    |
|           | Spearman $\rho$ | 0.633142    | 0.590657 | 0.629009 | 0.624100    |
| $h$       | Pearson $r$     | -0.001409   | 0.288386 | 0.464414 | 0.457842    |
|           | Spearman $\rho$ | 0.431172    | 0.342376 | 0.586917 | 0.613540    |

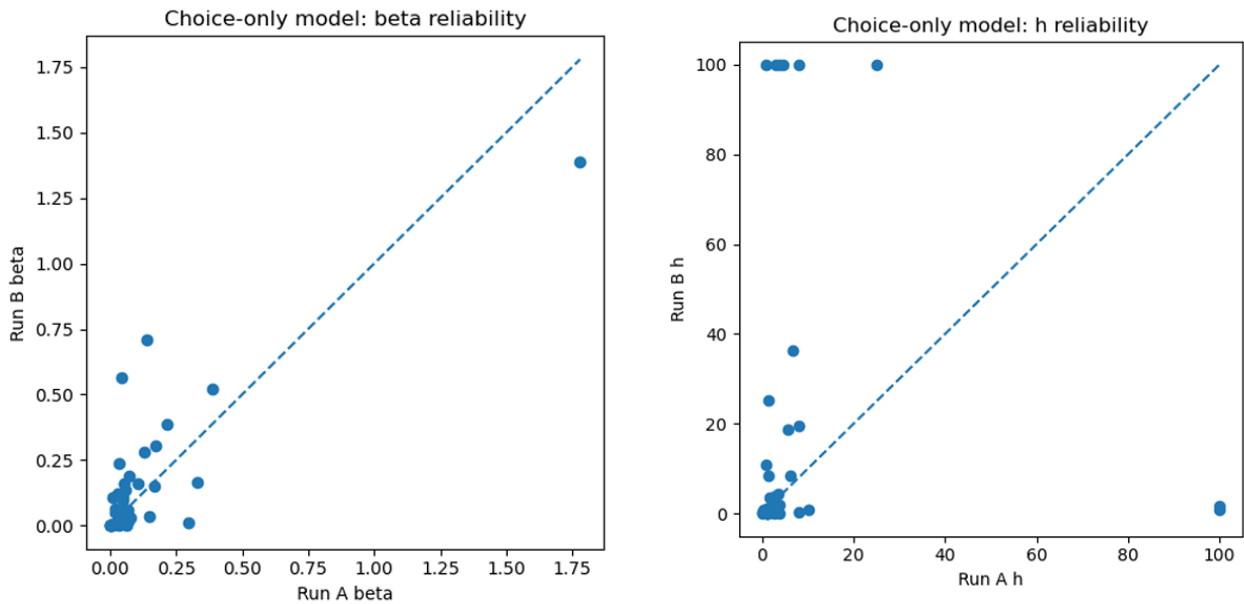


Figure 5: Choice-only reliability plots.

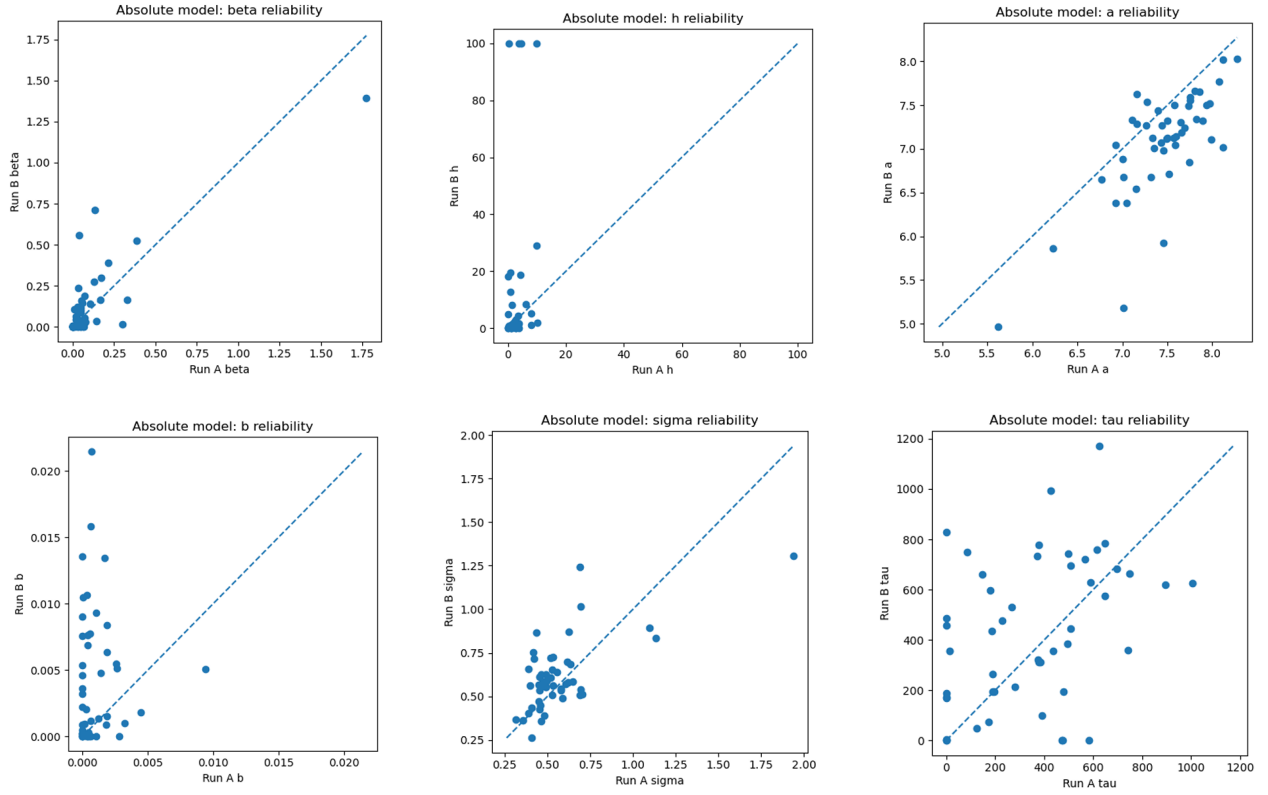


Figure 6: Absolute conflict model reliability plots.

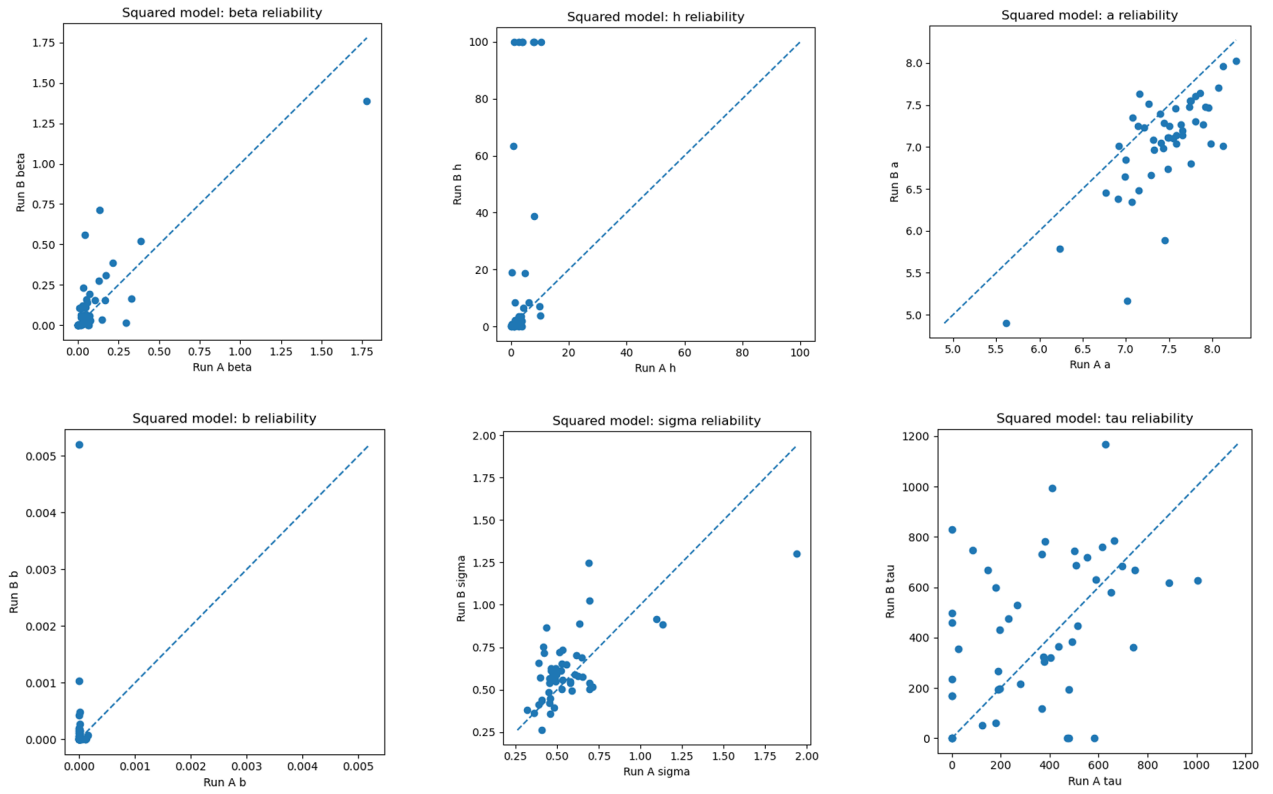


Figure 7: Squared conflict model reliability plots.

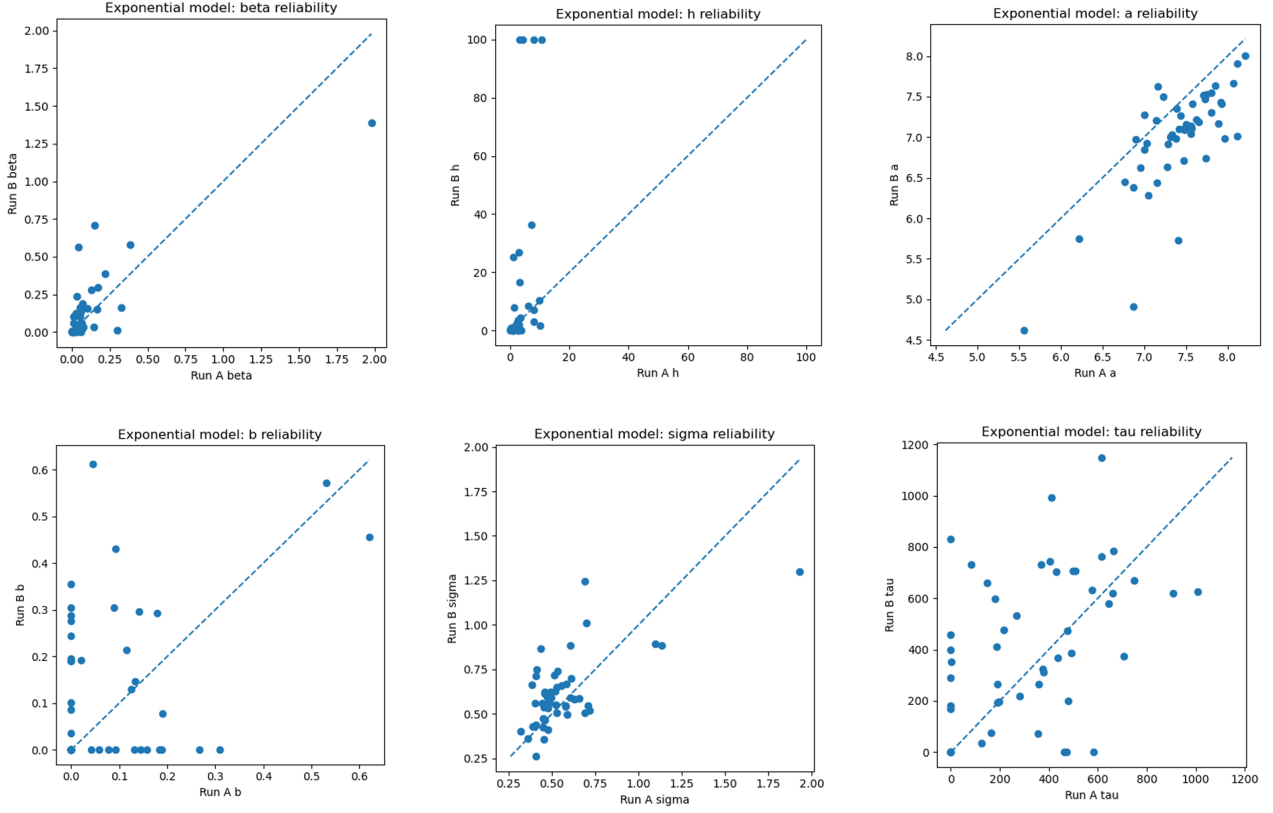


Figure 8: Exponential model reliability plots.

Table 7: Choice out-of-sample summary.

| Model       | Condition | Choice LL Mean | Choice LL SD | Choice Accuracy Mean | Choice Accuracy SD |
|-------------|-----------|----------------|--------------|----------------------|--------------------|
| Absolute    | Loss      | -0.662535      | 0.063997     | 0.649091             | 0.144825           |
|             | Reward    | -0.746081      | 0.691405     | 0.648590             | 0.218822           |
| Choice-only | Loss      | -0.670399      | 0.087960     | 0.631133             | 0.171363           |
|             | Reward    | -0.802461      | 0.770143     | 0.654304             | 0.223140           |
| Exponential | Loss      | -0.668214      | 0.083216     | 0.649501             | 0.150441           |
|             | Reward    | -0.852088      | 0.953404     | 0.663488             | 0.219511           |
| Squared     | Loss      | -0.662163      | 0.064141     | 0.641133             | 0.161402           |
|             | Reward    | -0.757446      | 0.723508     | 0.653896             | 0.216061           |

Table 8: Comparison of held-out choice predictive log-likelihood between each full RT model and the choice-only baseline. Holm correction was applied across the six model-condition comparisons.

| Comparison                  | Condition | $N$ | Mean Difference | Median Difference | Wilcoxon Statistic | Raw $p$  | $p_{\text{Holm}}$ |
|-----------------------------|-----------|-----|-----------------|-------------------|--------------------|----------|-------------------|
| Absolute vs. choice-only    | Reward    | 49  | 0.056380        | 0.000000          | 534.0              | 0.750891 | 1.000000          |
|                             | Loss      | 49  | 0.007864        | 0.000000012787    | 491.0              | 0.439820 | 1.000000          |
| Squared vs. choice-only     | Reward    | 49  | 0.045014        | -0.000000096035   | 485.0              | 0.403160 | 1.000000          |
|                             | Loss      | 49  | 0.008237        | 0.000000068918    | 475.0              | 0.346289 | 1.000000          |
| Exponential vs. choice-only | Reward    | 49  | -0.049627       | -0.000000016380   | 368.0              | 0.059480 | 0.356880          |
|                             | Loss      | 49  | 0.002186        | 0.000000005228    | 385.0              | 0.089339 | 0.446695          |

Table 9: Held-out RT predictive performance on common-support Run B trials. Panel A reports median participant-level predictive log-likelihood per trial. Panel B reports Friedman omnibus tests across the three RT mappings. Panel C summarizes Holm-corrected pairwise Wilcoxon comparisons.

**Panel A. Common-support RT predictive log-likelihood**

| Model             | Reward    | Loss      |
|-------------------|-----------|-----------|
| Absolute conflict | -8.294314 | -8.370851 |
| Squared conflict  | -8.315634 | -8.379263 |
| Exponential       | -8.252420 | -8.296339 |

**Panel B. Friedman omnibus tests**

| Condition | <i>N</i> | Friedman Statistic | <i>p</i> -value |
|-----------|----------|--------------------|-----------------|
| Reward    | 49       | 12.285714          | 0.002149        |
| Loss      | 49       | 7.877551           | 0.019472        |

**Panel C. Holm-corrected pairwise comparisons**

| Comparison               | Reward                   | Loss                     |
|--------------------------|--------------------------|--------------------------|
| Absolute vs. Squared     | n.s.                     | n.s.                     |
| Exponential vs. Absolute | $p_{\text{Holm}} < .001$ | $p_{\text{Holm}} = .024$ |
| Exponential vs. Squared  | $p_{\text{Holm}} = .001$ | $p_{\text{Holm}} = .015$ |

Table 10: RT out-of-sample MAE and RMSE metrics.

| Model             | Condition | MAE Mean   | MAE Median | MAE SD     | RMSE Mean   | RMSE Median |
|-------------------|-----------|------------|------------|------------|-------------|-------------|
| Absolute conflict | Loss      | 832.356769 | 816.454793 | 290.117540 | 1272.167257 | 1198.413964 |
|                   | Reward    | 830.864924 | 805.623087 | 343.051749 | 1275.398598 | 1161.135548 |
| Squared conflict  | Loss      | 832.207910 | 815.731728 | 287.087077 | 1271.505055 | 1196.519828 |
|                   | Reward    | 830.365714 | 800.256459 | 340.466670 | 1273.564394 | 1144.223700 |
| Exponential       | Loss      | 825.588932 | 797.258471 | 283.500436 | 1274.176525 | 1168.035253 |
|                   | Reward    | 835.814879 | 804.792577 | 349.859572 | 1271.247651 | 1161.135439 |

Table 11: Observed and predicted out-of-sample RT metrics and support coverage.

| Model             | Condition | Observed RT Mean | Predicted RT Median | Predicted RT Mean | Support Violations | Support Coverage |
|-------------------|-----------|------------------|---------------------|-------------------|--------------------|------------------|
| Absolute conflict | Loss      | 2060.173486      | 2214.892276         | 2613.787780       | 8                  | 99.84%           |
|                   | Reward    | 1854.191345      | 2175.514734         | 2568.969958       | 13                 | 99.73%           |
| Squared conflict  | Loss      | 2060.173486      | 2202.389641         | 2600.270464       | 8                  | 99.84%           |
|                   | Reward    | 1854.191345      | 2161.981245         | 2552.278175       | 13                 | 99.73%           |
| Exponential       | Loss      | 2060.173486      | 2207.917173         | 2608.932442       | 8                  | 99.84%           |
|                   | Reward    | 1854.191345      | 2218.213590         | 2607.299640       | 13                 | 99.73%           |

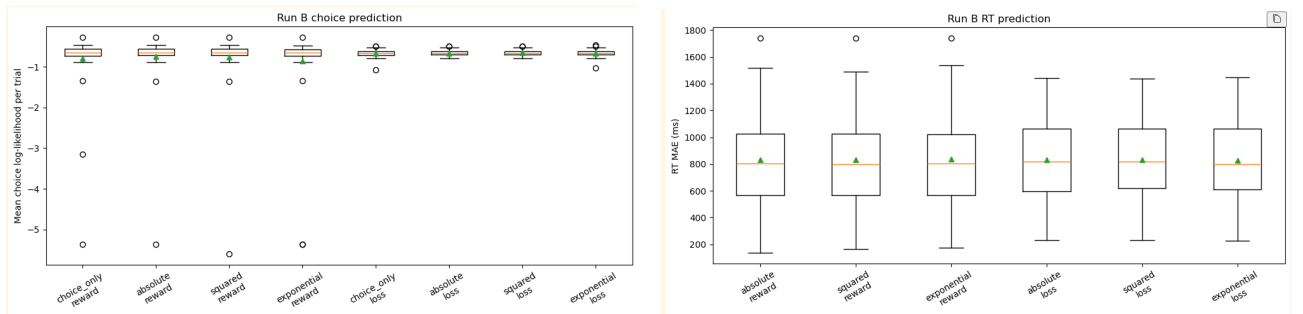


Figure 9: Choice and RT out-of-sample plots.

Table 12: Added value of RT relative to the choice-only baseline. Panel A summarizes held-out Run B choice prediction. Panel B compares recovery of the shared parameters  $\beta$  and  $h$ . Panel C summarizes test-retest reliability. For Panel A,  $\Delta LL = LL_{\text{full}} - LL_{\text{choice}}$ , such that positive values favor the full RT model. For Panel B, RMSE values are reported as full/choice-only, and positive median error improvement indicates lower absolute recovery error in the full model.

| <b>Panel A. Held-out choice prediction</b> |           |                  |                    |                          |           |  |
|--------------------------------------------|-----------|------------------|--------------------|--------------------------|-----------|--|
| Model                                      | Condition | Mean $\Delta LL$ | Median $\Delta LL$ | Median $\Delta$ Accuracy | Holm sig. |  |
| Absolute                                   | Reward    | 0.056            | 0.000              | 0.000                    | No        |  |
| Absolute                                   | Loss      | 0.008            | 0.000              | 0.000                    | No        |  |
| Squared                                    | Reward    | 0.045            | 0.000              | 0.000                    | No        |  |
| Squared                                    | Loss      | 0.008            | 0.000              | 0.000                    | No        |  |
| Exponential                                | Reward    | -0.050           | 0.000              | 0.000                    | No        |  |
| Exponential                                | Loss      | 0.002            | 0.000              | 0.000                    | No        |  |

| <b>Panel B. Recovery of shared parameters</b> |                  |               |                     |                        |
|-----------------------------------------------|------------------|---------------|---------------------|------------------------|
| Model                                         | $\beta$ RMSE F/C | $h$ RMSE F/C  | Full better for $h$ | Median $h$ improvement |
| Absolute                                      | 0.0287 / 0.0285  | 1.876 / 8.675 | 26/40               | 0.013                  |
| Squared                                       | 0.0303 / 0.0273  | 2.566 / 8.621 | 22/40               | 0.016                  |
| Exponential                                   | 0.0432 / 0.0435  | 8.197 / 8.240 | 25/40               | 0.006                  |

| <b>Panel C. Test-retest reliability</b> |                     |                 |                     |
|-----------------------------------------|---------------------|-----------------|---------------------|
| Model                                   | $\beta$ Pearson $r$ | $h$ Pearson $r$ | $h$ Spearman $\rho$ |
| Choice-only                             | 0.825               | -0.001          | 0.431               |
| Absolute                                | 0.826               | 0.288           | 0.342               |
| Squared                                 | 0.826               | 0.464           | 0.587               |
| Exponential                             | 0.823               | 0.458           | 0.614               |